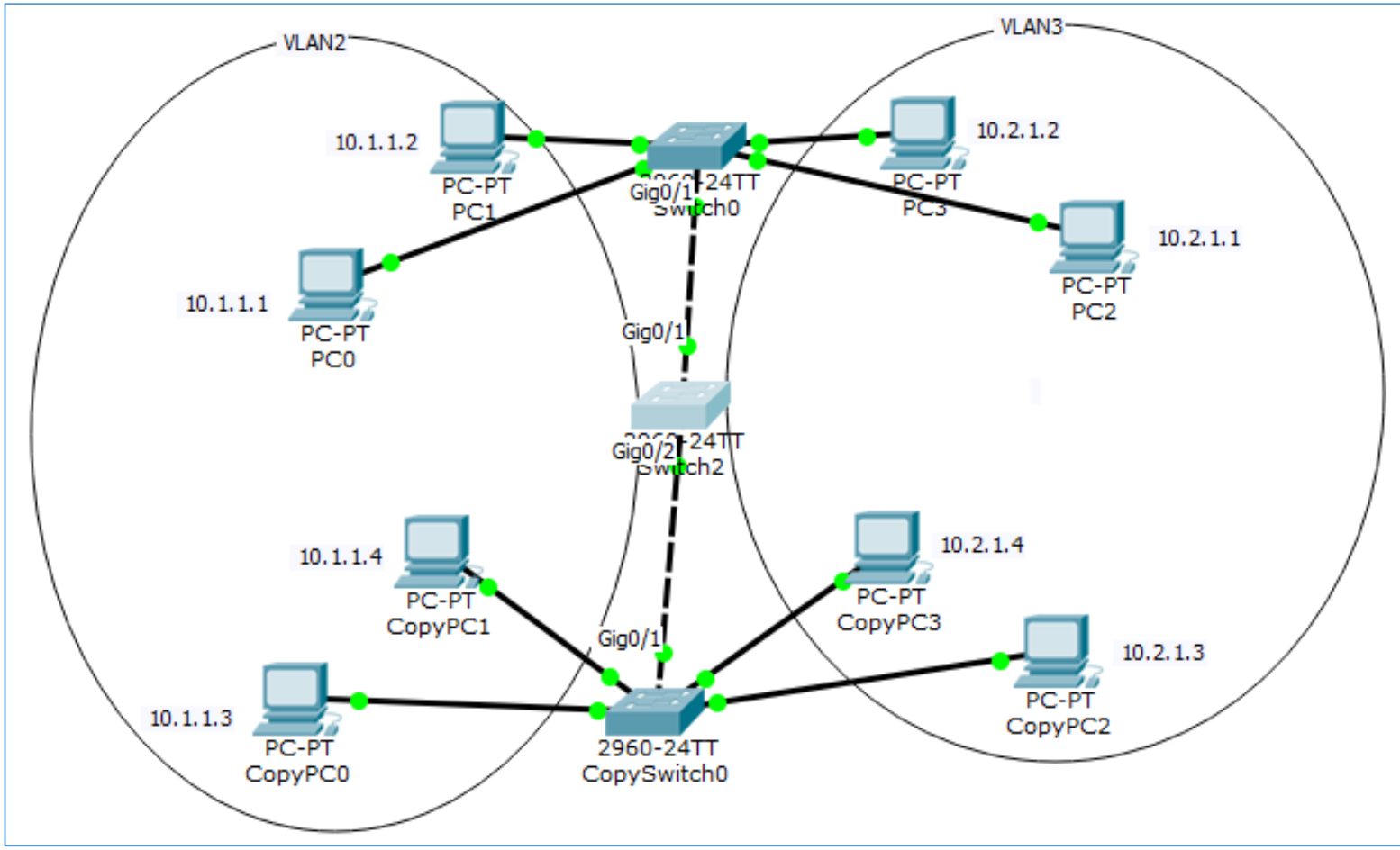
Network’s Lab.

Instructor: Masoud Asghari

Session: 8

S8_1.interVLan routing

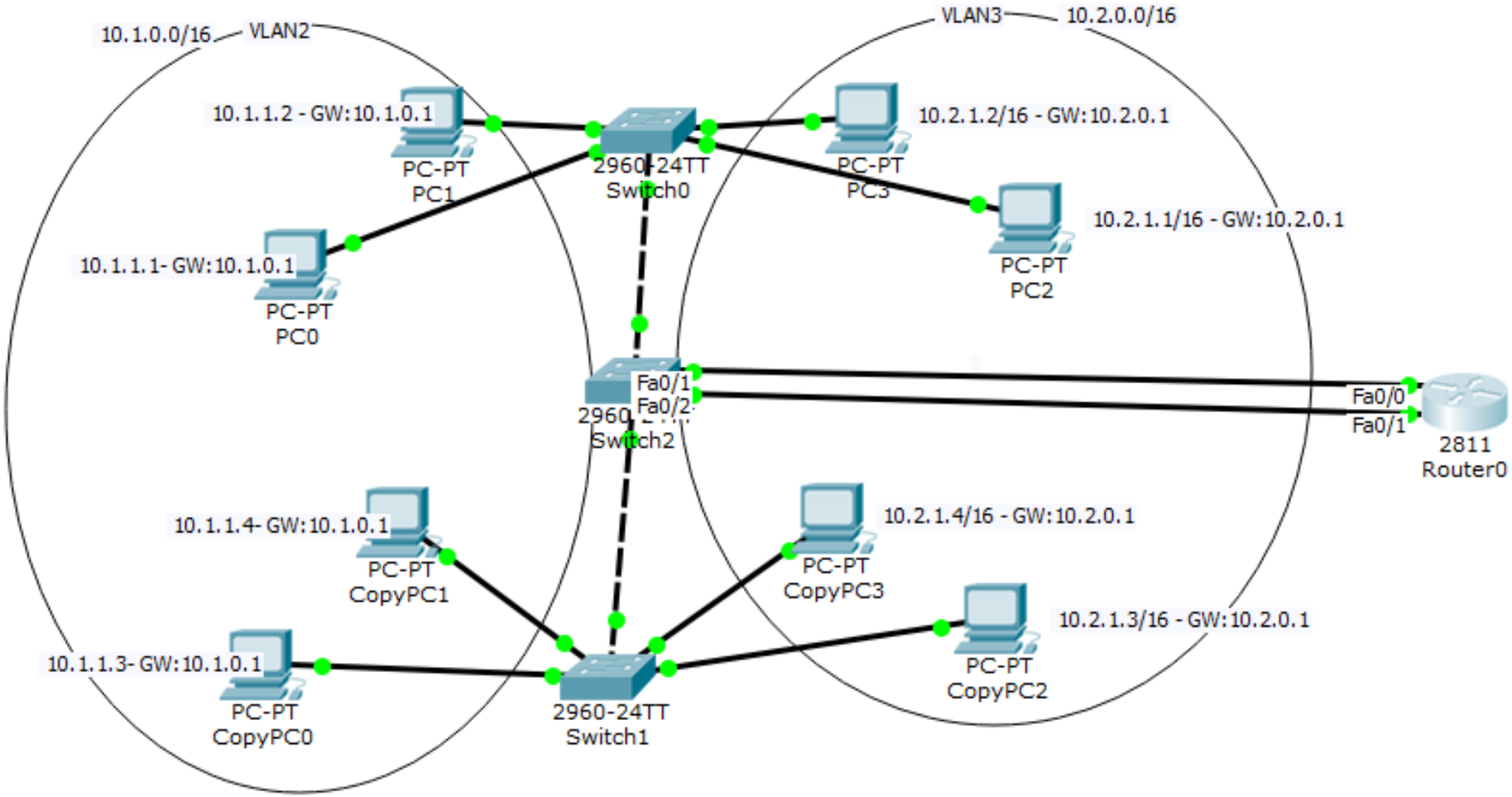
- Based on S6_1.



- We want to allow communications among VLans
  - Change **subnet mask** of all computers to 255.255.0.0
  - Add **default gateway** of 10.1.0.1 to all Vlan2 computers
  - Add **default gateway** of 10.2.0.1 to all Vlan3 computers

S8_1.interVLan routing

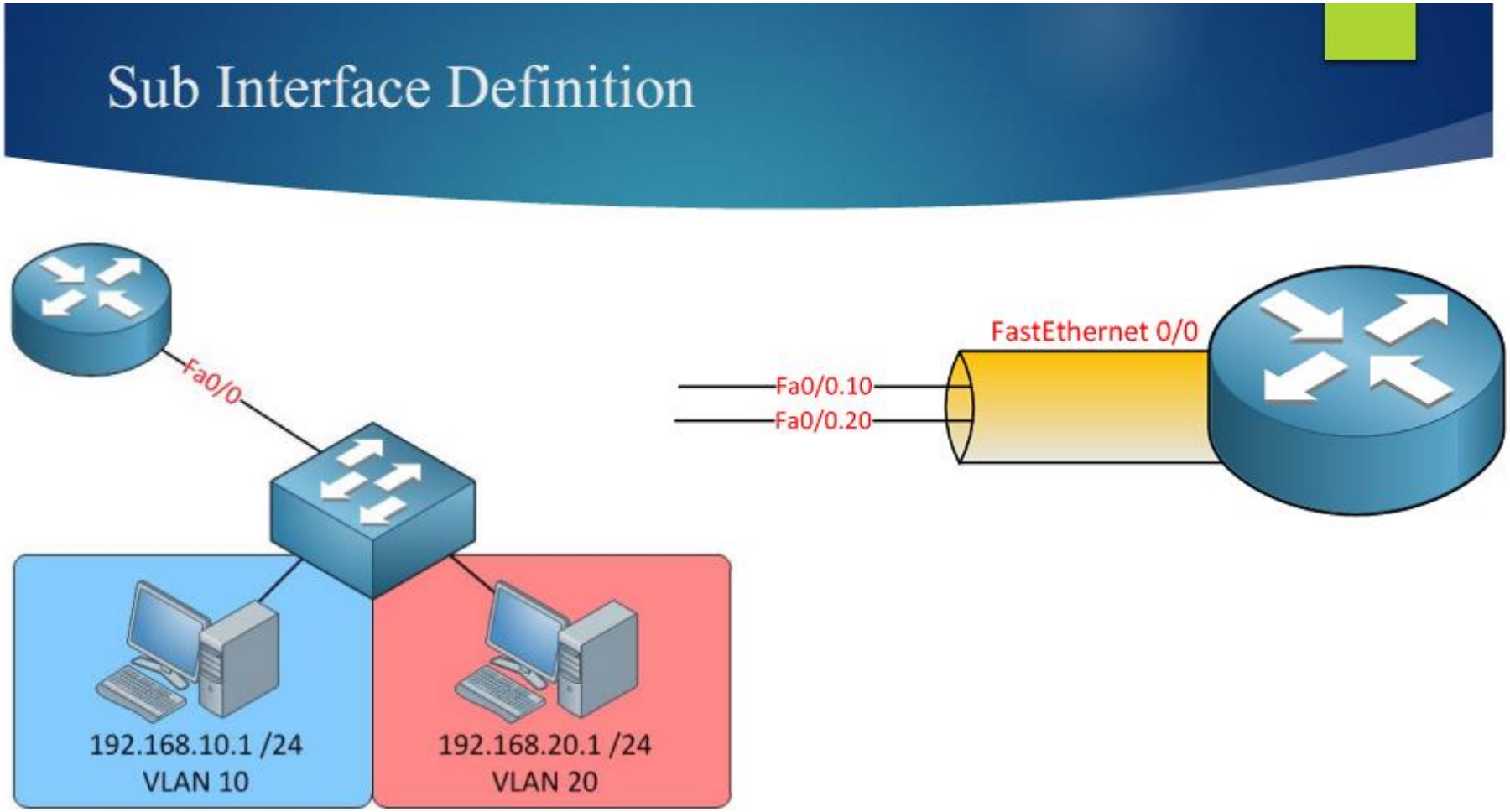
- Add a router



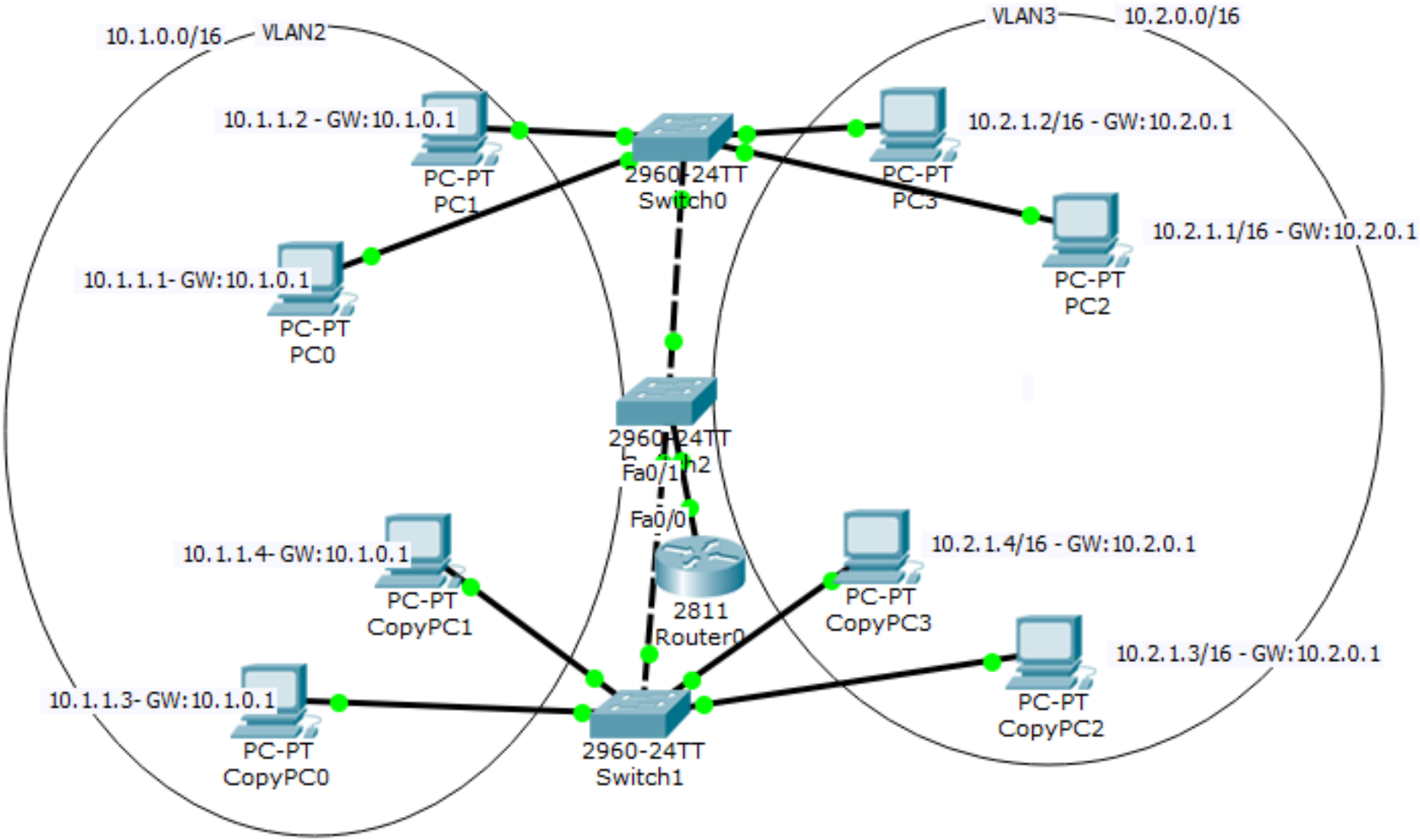
S8_1.interVLan routing

- On Switch 2 (the middle switch)
  - Switch2(config)#interface fastEthernet 0/1
  - Switch2(config-if)#switchport mode access
  - Switch2(config-if)#switchport access vlan 2
  - Switch2(config)#interface fastEthernet 0/2
  - Switch2(config-if)#switchport mode access
  - Switch2(config-if)#switchport access vlan 3
- On the router
  - Router(config)#interface fastEthernet 0/0
  - Router(config-if)#ip address **10.1.0.1** 255.255.0.0
  - Router(config-if)#no shutdown
  - Router(config-if)#exit
  - Router(config)#interface fastEthernet 0/1
  - Router(config-if)#ip address **10.2.0.1** 255.255.0.0
  - Router(config-if)#no shutdown

Sub Interface



S8_2.InterVLan routing (sub-int.), Based on **S8_1**



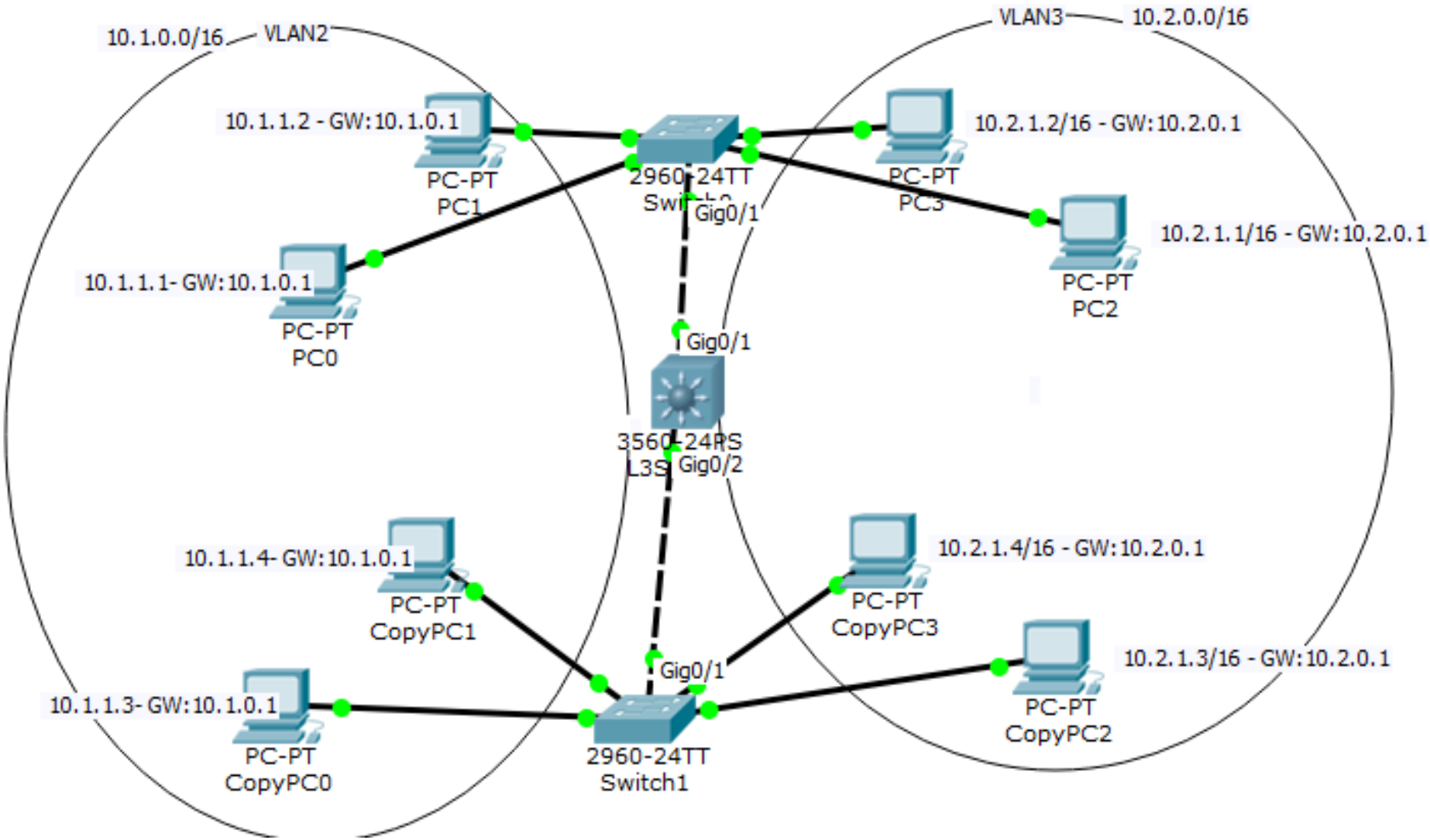
S8_2.InterVLan routing(sub-int.)

- On Switch 2
  - Switch2(config)#interface fastEthernet 0/1
  - Switch2(config-if)#switchport mode trunk

- On router
  - Router(config)#interface fastEthernet 0/0
  - Router(config-if)#no shutdown
  - Router(config-if)#**no ip address**
  - Router(config)#interface fastEthernet 0/1
  - Router(config-if)#shutdown
  - Router(config-if)#**no ip address**
  - Router(config)#interface fastEthernet 0/0.1
  - Router(config-subif)#encapsulation dot1Q **2**
  - Router(config-subif)#ip address 10.1.0.1 255.255.0.0
  - Router(config-subif)#exit
  - Router(config)#interface fastEthernet 0/0.2
  - Router(config-subif)#encapsulation dot1Q **3**
  - Router(config-subif)#ip address 10.2.0.1 255.255.0.0
  - Router(config-subif)#exit

Was Set at S8_1

S8_3.InterVLan routing (L3Switch.), Based on S8_2



S7_3.interVLan routing (L3Switch.)

- L3Switch#show interfaces trunk
- L3Switch#conf t
- L3Switch(config)#vlan 2
- L3Switch(config-vlan)#exit
- L3Switch(config)#vlan 3
- L3Switch(config-vlan)#exit
- L3Switch(config)#interface vlan 2
- L3Switch(config-if)#ip address 10.1.0.1 255.255.0.0
- L3Switch(config-if)#exit
- L3Switch(config)#interface vlan 3
- L3Switch(config-if)#ip address 10.2.0.1 255.255.0.0
- L3Switch(config-if)#exit
- L3Switch(config)#**ip routing**

L3Switch(config)#interface gigabitEthernet 0/1
L3Switch(config-if)#**switchport trunk encapsulation dot1q**
L3Switch(config-if)#**switchport mode trunk**
L3Switch(config)#interface gigabitEthernet 0/2
L3Switch(config-if)#switchport trunk encapsulation dot1q
L3Switch(config-if)#switchport mode trunk

Assignments

- Find out what is the similarities and differences between a *router* and a *Layer3 switch*.
  - Bring it in a paper